South African Provincial & Metropolitan Residential Property Market Analytics

Project plan outlining the objective, scope, analytical workflow, datasets, technology stack and expected outcomes.

1. Project Objective

Analyse South African residential property price movements across provinces and metropolitan areas using historical Stats SA data. The project aims to identify regional market trends, measure property price growth and volatility, evaluate market stability, and forecast residential property price movements over the next 12 months.

2. Scope

- Data extraction and preparation from Stats SA residential property price index data.
- Data cleaning, transformation and standardisation using Python.
- Separate datasets for provincial, metropolitan and overall metropolitan property price indices.
- Feature engineering, including monthly growth, annual growth, long-term growth and volatility.
- Regional property market performance and stability analysis.
- Time-series forecasting and model evaluation.
- Interactive Power BI dashboards for reporting and analysis.
- Project documentation and version control using GitHub.

3. Architecture Overview

Stats SA Data → Python Data Processing → Feature Engineering → Statistical Analysis & Forecasting → Excel Output Datasets → Power BI Dashboards

Component	Purpose
Source Data	Original residential property price index data from Stats SA
Data Processing	Clean, restructure and separate provincial, metropolitan and overall metropolitan datasets
Feature Engineering	Calculate monthly growth, annual growth, long-term growth and volatility
Statistical Analysis	Evaluate regional performance, growth patterns and market stability
Forecasting	Generate 12-month forecasts and evaluate model performance
Output Datasets	Store processed, feature-engineered and forecast results
Power BI	Present market trends, comparisons, risk analysis and forecasts

4. Project Phases

Phase	Activities
Planning	Defined project objectives, analytical requirements and scope
Data Understanding	Examined Stats SA residential property price index data and identified relevant fields
Data Preparation	Cleaned, reshaped and separated provincial, metropolitan and overall metropolitan datasets
Feature Engineering	Developed Python calculations for monthly growth, annual growth, long-term growth and volatility

Phase	Activities
Exploratory Analysis	Analysed historical property price movements and regional performance
Forecasting	Applied Holt-Winters Exponential Smoothing to generate 12-month forecasts
Model Evaluation	Evaluated forecast performance using test data and error metrics
Reporting	Developed Power BI dashboards to communicate market trends, performance, stability and forecasts
Documentation & Deployment	Organised Python scripts, datasets, documentation and project resources in GitHub

5. Main Datasets

Provincial Datasets

- Residential_Property_Price_Province.xlsx
- Residential_Property_Price_Province_Features.xlsx

Metropolitan Datasets

- Residential_Property_Price_Metro.xlsx
- Residential_Property_Price_Metro_Features.xlsx
- Residential_Property_Price_Metro_Overall.xlsx
- Residential_Property_Price_Metro_Overall_Features.xlsx

Forecasting Outputs

- Province_Property_Price_Forecast.xlsx
- Metro_Property_Type_Forecast.xlsx
- Metro_Overall_Forecast.xlsx

Key Analytical Measures

- Monthly Growth (MoM), Year-on-Year Growth (YoY), Long-Term Growth and Price Growth Volatility
- Forecasted Property Price Index and Forecast Model Performance

6. Technology Stack

Component	Tool
Python Development & Data Processing	Visual Studio
Programming Language	Python
Data Manipulation	Pandas, NumPy
Statistical Analysis & Forecasting	Statsmodels
Data Storage & Output Files	Microsoft Excel
Data Visualisation & Reporting	Microsoft Power BI
Version Control & Project Hosting	GitHub

7. Deliverables

- Cleaned and transformed provincial and metropolitan datasets.
- Python data transformation and feature engineering scripts.
- Historical property market analysis and calculated performance measures.
- 12-month forecasting outputs.
- Forecast model evaluation results.
- Interactive Power BI dashboards.
- Project architecture and data flow documentation.
- GitHub repository containing project code, datasets and documentation.

8. Expected Outcome

A complete residential property market analytics solution that transforms historical Stats SA data into meaningful market insights. The solution enables users to compare provincial and metropolitan property market performance, understand growth and volatility patterns, assess regional market stability, and examine expected property price movements over the next 12 months.